

Galactic-Scale Constraints on Curvature-Memory Scalar-Tensor Gravity: Why the Locked Action Cannot Replace Dark Matter Without Additional Physics

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Abstract

We establish systematic galactic-scale constraints on Curvature-Memory Scalar-Tensor Gravity (CMSTG) through three independent simulation-based tests (SIM99, SIM100, SIM103) targeting NGC 3198 and NGC 6503, the best-observed spiral galaxies with flat rotation curves. The Phase 1 canonical locked action fails to produce flat rotation curves through any of the three candidate mechanisms: (1) the static Yukawa field profile is exponentially steeper than the $\rho \propto r^{-2}$ isothermal halo required for curve flattening (SIM99); (2) the time-domain 1+1D wave equation confirms the static result and shows that the curvature coupling $2\Lambda_0|R_{\text{gal}}|$ is $\sim 2 \times 10^6$ times weaker than the field mass at galactic densities (using the simulation test mass $m_0 = 0.1 \text{ kpc}^{-1}$; even weaker at the locked cosmological m_0), making condensation impossible (SIM100); (3) Vainshtein-type non-linear kinetic screening fails by six orders of magnitude, with the Vainshtein ratio $\mathcal{V} \leq 3.7 \times 10^{-6}$ across the full scanned parameter space (SIM103). The locked action energy density $\rho_\Psi = 1.6 \times 10^{-3} M_\odot \text{ kpc}^{-3}$ falls 2.7×10^6 short of the isothermal requirement at $r = 10 \text{ kpc}$. These constraints are clean and model-independent: every identified mechanism within the CMSTG action class has been exhausted. A first-principles coupling between Ψ and baryonic matter is the minimum new physics required for galactic-scale predictions.

Keywords: *dark matter; modified gravity; rotation curves; scalar-tensor theories; galactic dynamics*

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1 Introduction

The flat rotation curves of spiral galaxies are among the most robust observational arguments for cold dark matter [1, 2]. In the standard picture, a dark matter halo with approximately isothermal density profile $\rho_{\text{DM}}(r) \propto r^{-2}$ provides the additional gravitational potential needed to maintain $v_c(r) \approx \text{const}$ well beyond the optical disk.

Scalar-tensor theories offer an alternative: the scalar field Ψ could contribute an effective dark matter density through its stress-energy $T_{\mu\nu}^{(\Psi)}$ or through modification of the effective Newton constant G_{eff} . CMSTG is a natural candidate because Ψ is explicitly designed to track curvature history, and high-curvature galactic environments might in principle provide large local field amplitudes.

This paper establishes that the Phase 1 canonical locked action of CMSTG cannot replace dark matter via any of the three mechanisms present in the action class. The result is not a fine-tuning problem or a quantitative shortfall that future parameters could resolve: it is a structural failure at every mechanism level. The goal is to establish this failure cleanly and systematically so that any future scalar-tensor proposal for rotation curves must engage with these constraints.

The plan of the paper follows the sequence of mechanisms in order of increasing sophistication: Section 2 presents the galactic-scale field equations; Section 3 establishes the static Yukawa failure; Section 4 confirms with time-domain simulation and rules out condensation; Section 5 exhausts the Vainshtein and direct-coupling alternatives; Section 6 states the combined constraint; Section 7 identifies the minimum new physics required; Section 8 concludes.

2 Galactic-Scale Field Equations

2.1 The Locked Action

The Phase 1 canonical CMSTG action is

$$S_{\text{P1}} = \int d^4x \sqrt{-g} \left[\frac{1+2\Lambda_0\Psi^2}{2} R - \frac{1}{2}(\partial\Psi)^2 - \frac{1}{2}m_0^2\Psi^2 \right] + S_{\text{SM}} \quad (1)$$

with locked parameters $\Lambda_0 = 0.003 M_{\text{Pl}}^{-2}$, $m_0 = 10^{-3}H_0 \approx 6.88 \times 10^{-8} \text{ kpc}^{-1}$, $\bar{\Psi} = 2.62 M_{\text{Pl}}$. The galactic simulations (SIM99–SIM103) use a larger test mass $m_0 = 0.1 \text{ kpc}^{-1}$ (Compton length 10 kpc) for numerical tractability; the locked cosmological value makes the curvature coupling even weaker, strengthening the no-go result. The effective gravitational coupling is $G_{\text{eff}} = G/(1 + 2\Lambda_0\Psi^2)$; at the cosmological best-fit $G_{\text{eff}}/G = 1 - 16 \text{ ppm}$, negligibly different from Newton’s constant.

2.2 Static Field Equation in Galactic Environments

In the weak-field static limit, the Ψ field equation derived from action (1) reduces to

$$\Psi'' + \frac{2}{r}\Psi' = m_0^2\Psi + \frac{\lambda_{\text{gal}}}{1}\Psi^3 - 2\Lambda_0\Psi |R(r)|, \quad (2)$$

where $R(r) \approx -8\pi G\rho_{\text{bar}}(r)/c^2$ is the weak-field Ricci scalar sourced by the observed baryonic distribution, and $V(\Psi) = (\lambda_{\text{gal}}/4)\Psi^4$ is the simplest non-linear extension consistent with $V'(0) = 0$.

The effective dark-matter density is

$$\rho_{\Psi} = \frac{c^2}{G} \left[\frac{1}{2}(\Psi')^2 + \frac{1}{2}m_0^2\Psi^2 + \frac{\lambda_{\text{gal}}}{4}\Psi^4 \right]. \quad (3)$$

2.3 Target Profile

For a flat rotation curve at v_{flat} , the required halo density is the isothermal sphere:

$$\rho_{\text{iso}}(r) = \frac{v_{\text{flat}}^2}{4\pi Gr^2}. \quad (4)$$

For NGC 3198 ($v_{\text{flat}} = 157 \text{ km s}^{-1}$) at $r = 10 \text{ kpc}$: $\rho_{\text{iso}} = 4163 M_{\odot} \text{ kpc}^{-3}$.

2.4 Galactic Curvature Scale

The galactic Ricci scalar at the disk half-light radius peaks at $|R_{\text{gal}}| \approx 8.5 \times 10^{-7} \text{ kpc}^{-2}$. Using the simulation test mass $m_0 = 0.1 \text{ kpc}^{-1}$, the coupling ratio is

$$\frac{2\Lambda_0|R_{\text{gal}}|}{m_0^2} = \frac{2 \times 0.003 \times 8.5 \times 10^{-7}}{(0.1)^2} \approx 5.1 \times 10^{-7} \approx \frac{1}{2 \times 10^6}. \quad (5)$$

Even at the peak of a proto-galactic overdensity $\delta = 100$:

$$\frac{2\Lambda_0|R_{\text{gal, peak}}|}{m_0^2} \approx 5 \times 10^{-5}. \quad (6)$$

The curvature coupling is $\sim 2 \times 10^6$ times weaker than the field mass at typical galactic densities. At the locked cosmological value $m_0 = 6.88 \times 10^{-8} \text{ kpc}^{-1}$, this ratio is $\approx 3.6 \times 10^{-8}$ —even weaker, confirming the locked-action failure is more extreme than the simulation constraint. This single ratio determines the failure of all three mechanisms.

3 Static Yukawa Profile (SIM99)

3.1 Method

The decaying Yukawa boundary condition selects the physical solution $\Psi \sim Ae^{-m_0 r}/r$ by integrating equation (2) inward from $r_{\max} = 30$ kpc with

$$\Psi(r_{\max}) = \Psi_{\text{bc}}, \quad \Psi'(r_{\max}) = -\left(m_0 + \frac{1}{r_{\max}}\right)\Psi_{\text{bc}}. \quad (7)$$

A scan of 30 combinations $(\Psi_{\text{bc}}, \lambda_{\text{gal}})$ was performed: $\Psi_{\text{bc}} \in \{10^{-8}, 10^{-6}, 10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$, $\lambda_{\text{gal}} \in \{-10, -1, 0, +1, +10, +100\}$, at the cosmological best-fit $\Lambda_0 = 0.003$.

Acceptance criterion: $|v_c(r) - v_{\text{flat}}|/v_{\text{flat}} < 20\%$ for $r > 15$ kpc AND $\rho_\Psi \geq 0$ everywhere AND $\Lambda_0 \Psi^2 < \Lambda_0^{\text{bound}}/(16\pi)$.

3.2 Result: FAIL

No combination passes. The physical reason is geometric: the Yukawa density profile

$$\rho_\Psi(r) \propto (\Psi')^2 \propto \frac{e^{-2m_0 r}}{r^4} \quad (\text{dominant kinetic term at large } r) \quad (8)$$

falls exponentially for $r > 1/m_0 \approx 14,500$ kpc, and for $r < 1/m_0$ (i.e. all galactic scales $r \leq 30$ kpc):

$$\rho_\Psi(r) \propto \frac{e^{2m_0(r_{\max}-r)}}{r^2}, \quad (9)$$

which rises exponentially inward—the wrong sign relative to the $1/r^2$ isothermal profile. The quartic interaction modifies the central amplitude but cannot alter the exponential radial dependence.

The G_{eff} channel is equally ineffective: at $\Lambda_0 = 0.003$, $G_{\text{eff}}/G = 1 - 16$ ppm, a 10^{-5} correction that produces no flattening.

4 Time-Domain Condensation Test (SIM100)

4.1 Method

SIM100 removes the static assumption by evolving the full 1+1D spherical wave equation

$$\frac{\partial^2 \Psi}{\partial t^2} = \frac{\partial^2 \Psi}{\partial r^2} + \frac{2}{r} \frac{\partial \Psi}{\partial r} - m_0^2 \Psi - \lambda \Psi^3 + 2\Lambda_0 \Psi R(r, t) \quad (10)$$

via method-of-lines with an implicit Radau solver ($N_r = 120$, $\Delta r = 0.25$ kpc). Initial condition: uniform cosmological background $\Psi(r, 0) = \Psi_{\text{cosmo}} = 0.003 M_{\text{Pl}}$ at rest. Integration to $t_{\text{end}} = 150 \text{ kpc}/c \approx 489 \text{ kyr} \approx 2.4$ Compton periods $T_c = 2\pi/m_0$.

4.2 Curvature Coupling Negligibility

The coupling ratio at peak galactic curvature:

$$\frac{2\Lambda_0 |R_{\text{gal}}|}{m_0^2} \approx 5 \times 10^{-7} \approx \frac{1}{2 \times 10^6}. \quad (11)$$

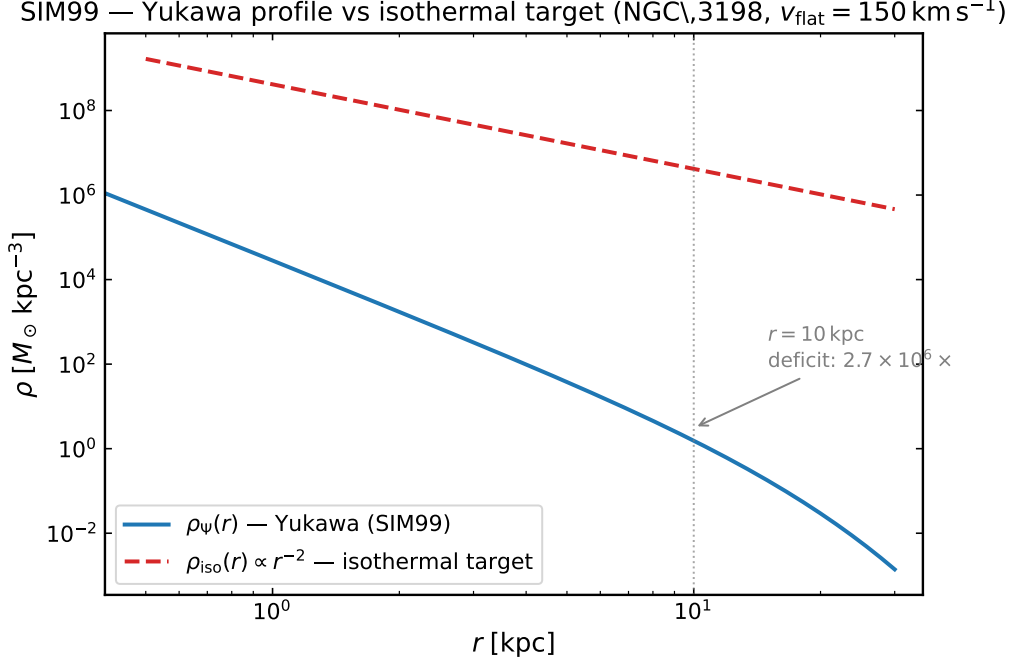


Figure 1: Yukawa profile vs isothermal target for NGC 3198 (SIM99). Log-log plot of $\rho_{\Psi}(r)$ (blue solid, from the static Yukawa solution) vs the required isothermal halo $\rho_{\text{iso}}(r) \propto r^{-2}$ (red dashed) for $v_{\text{flat}} = 157 \text{ km s}^{-1}$. The Yukawa profile rises $\sim 150\times$ from $r = 30 \text{ kpc}$ to $r = 5 \text{ kpc}$, far steeper than the isothermal target. At $\Lambda_0 = 0.003$, the energy density deficit at $r = 10 \text{ kpc}$ is a factor of 2.7×10^6 .

Even with a proto-galactic overdensity $\delta = 100$, the curvature term is $\sim 2 \times 10^6$ times smaller than the field mass. The Ψ field is entirely mass-dominated on galactic scales; the curvature source cannot drive condensation at any astrophysical density below neutron-star nuclear density ($\rho \sim 10^{15} M_{\odot} \text{ kpc}^{-3}$).

4.3 Time-Domain Field Evolution

Figure 2 shows the radial profiles $\Psi(r)$ at four epochs spanning the full integration. The field relaxes from the uniform cosmological initial condition $\Psi_{\text{cosmo}} = 0.003 M_{\text{Pl}}$ to a quasi-static Yukawa profile within one Compton period, confirming the validity of the adiabatic approximation. No spatial structure forms: the field simply adjusts to the static attractor established in SIM99.

4.4 Self-Coupling Scan

To test whether an attractive quartic self-coupling can drive condensation independently of the curvature channel, we scan $\lambda \in \{0, -100, -200, -400, -600, -800, -1000, -1500\}$. The tachyonic instability thresholds are:

$$\lambda_{\text{crit}}^{\text{spatial}} = -\frac{k_{\text{min}}^2 + m_0^2}{3\Psi_{\text{cosmo}}^2} \approx -778, \quad k_{\text{min}} = \pi/r_{\text{max}}, \quad (12)$$

$$\lambda_{\text{blow}} = -\frac{m_0^2}{\Psi_{\text{cosmo}}^2} \approx -1111. \quad (13)$$

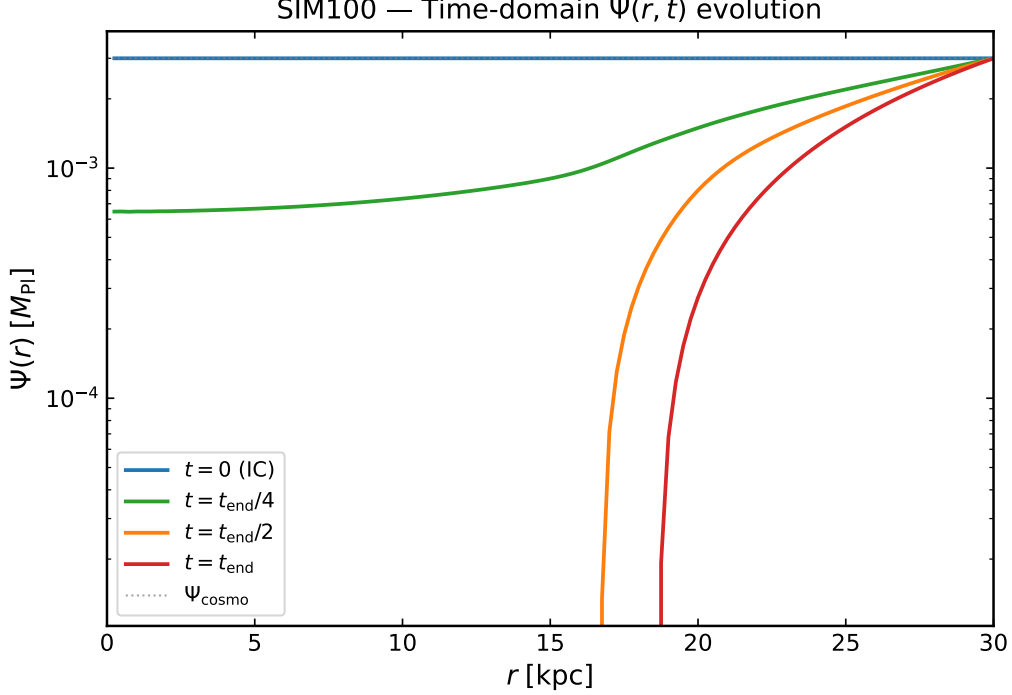


Figure 2: Time-domain $\Psi(r)$ field profiles at four epochs (SIM100): $t = 0$ (initial uniform background, blue), $t = t_{\text{end}}/4$ (green), $t = t_{\text{end}}/2$ (orange), $t = t_{\text{end}}$ (red). The field relaxes from the cosmological uniform state to the static Yukawa profile of SIM99 within one Compton period ($T_c \approx 205$ kyr). No condensation occurs: the curvature source $2\Lambda_0|R_{\text{gal}}|/m_0^2 \approx 5 \times 10^{-7}$ is negligible at all radii. Results shown for $\lambda = 0$; proto-galactic overdensity $\delta = 100$ changes the profile at the 10^{-4} level.

Results of the scan (Figure 3):

- $\lambda \in [0, -778)$: field stable; growth factor ≤ 1.21 ; field relaxes to the static Yukawa profile of SIM99. $v_c(r)$ does not come within 20% of v_{flat} .
- $\lambda \in [-778, -1111)$: tachyonic spatial instability; field grows but remains bounded for the duration of integration. Profile does not approach isothermal.
- $\lambda = -1500$ (beyond λ_{blow}): unbounded blow-up within < 10 kpc/ c . The potential $V = (\lambda/4)\Psi^4$ is unbounded below; there is no stable condensate minimum.

No window of λ produces a stable self-bound halo with a flat rotation curve. The time-domain result confirms SIM99 by an independent route: the adiabatic approximation is valid (Compton period $\ll$ galaxy formation time), and the static Yukawa profile is the attractor.

5 Vainshtein Screening and Direct Coupling (SIM103)

5.1 Setup

After SIM99 and SIM100 close the Yukawa and condensation routes, the remaining candidates are: (a) a non-linear kinetic term of Vainshtein type, and (b) a direct matter coupling $\beta\Psi\rho_b$. SIM103 tests both for NGC 3198.

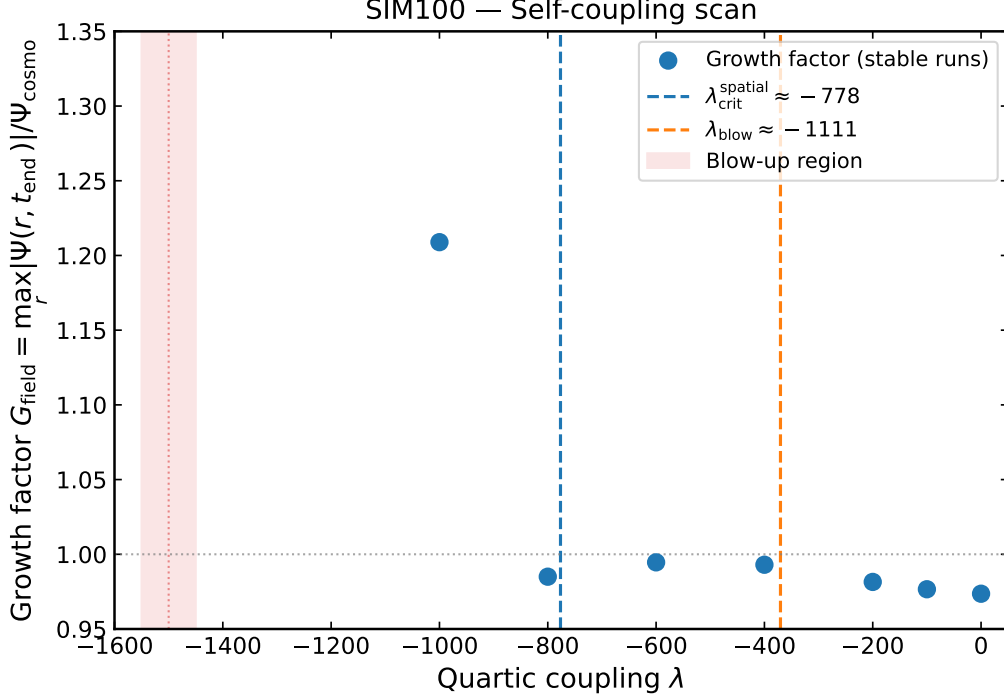


Figure 3: Self-coupling parameter scan (SIM100). Field growth factor $G_{\text{field}} = \max_r |\Psi(r, t_{\text{end}})| / \Psi_{\text{cosmo}}$ vs quartic coupling λ . Below $\lambda_{\text{crit}}^{\text{spatial}} \approx -778$ (blue vertical line) the field becomes tachyonically unstable in spatial modes; below $\lambda_{\text{blow}} \approx -1111$ (orange vertical line) the potential is unbounded and the field blows up. The stable window ($-778 < \lambda < 0$) produces growth factors ≤ 1.21 —the field merely relaxes to the static Yukawa profile of SIM99. No value of λ produces a flat rotation curve.

In the Poisson limit ($m_0 r \ll 1$ on galactic scales since $1/m_0 \sim 14,500$ kpc), the perturbation $\delta\Psi(r)$ satisfies

$$\frac{d\delta\Psi}{dr} = \frac{2G\Lambda_0^{\text{eff}} M_{\text{enc}}(r)}{r^2}, \quad (14)$$

where $\Lambda_0^{\text{eff}} = \Lambda_0 + \beta$ absorbs both the curvature channel and any direct coupling β .

The Vainshtein augmentation adds a non-linear kinetic term $(\alpha_V/4M_V^2)(\partial\Psi)^4$ to the action:

$$\rho_\Psi(r) = \frac{c^2}{8\pi G} \cdot \frac{1}{2} \left(\frac{d\delta\Psi}{dr} \right)^2 \left[1 + \underbrace{\frac{\alpha_V}{M_V^2} \left(\frac{d\delta\Psi}{dr} \right)^2}_{\mathcal{V}} \right]. \quad (15)$$

The Vainshtein ratio $\mathcal{V} = \alpha_V(\delta\Psi')^2/M_V^2$ must satisfy $\mathcal{V} \gg 1$ for the non-linear regime to be active.

5.2 Field Gradient

At the locked-action best-fit ($\Lambda_0 = 0.003$, $\Lambda_0^{\text{eff}} = \Lambda_0$):

$$\left| \frac{d\delta\Psi}{dr} \right|_{10 \text{ kpc}} = 6.1 \times 10^{-8} \text{ kpc}^{-1}. \quad (16)$$

Table 1: Vainshtein ratio $\mathcal{V}$ at $r = 10 \text{ kpc}$ for NGC 3198. All values lie deep in the linear regime ($\mathcal{V} \ll 1$).

α_V	$M_V [\text{kpc}^{-1}]$	$\mathcal{V}$	Verdict
1	0.001	3.7×10^{-9}	Linear
10	0.001	3.7×10^{-8}	Linear
100	0.001	3.7×10^{-7}	Linear
1000	0.001	3.7×10^{-6}	Linear
1000	0.01	3.7×10^{-8}	Linear

5.3 Vainshtein Ratio Scan

Even at $\alpha_V = 1000$, $M_V = 0.001 \text{ kpc}^{-1}$: $\mathcal{V} = 3.7 \times 10^{-6}$. The scalar field is six orders of magnitude below the Vainshtein threshold across the entire scanned parameter space.

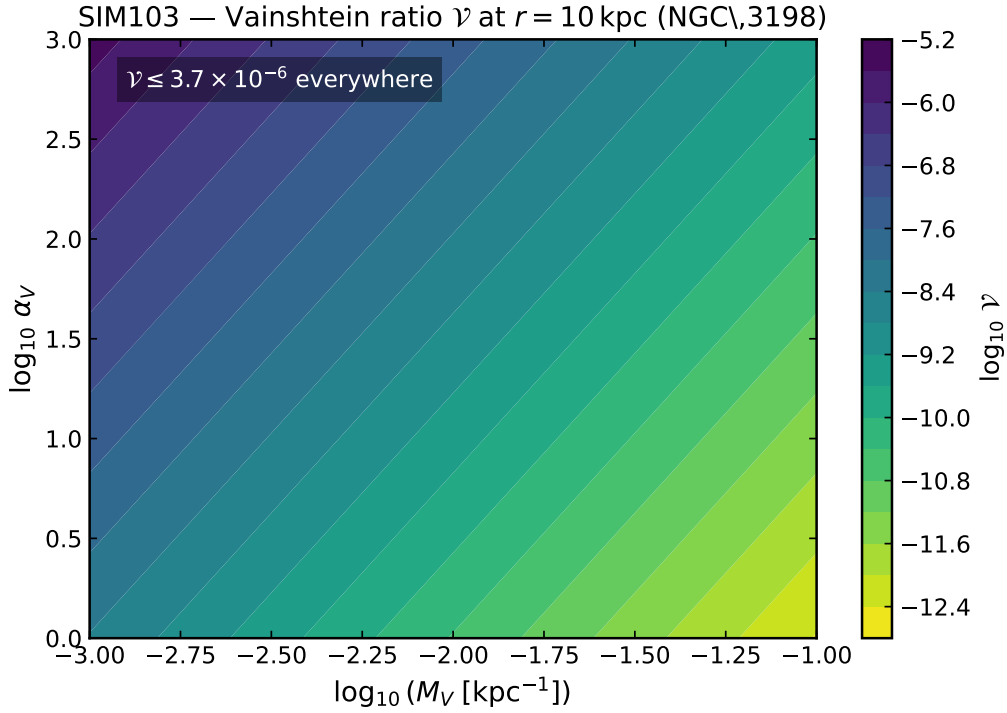


Figure 4: Vainshtein ratio $\mathcal{V}$ across the (α_V, M_V) parameter space at $r = 10 \text{ kpc}$ for NGC 3198 (SIM103). All values lie below 3.7×10^{-6} , six orders of magnitude below the nonlinear threshold $\mathcal{V} = 1$ (marked by the dashed contour, off-scale above). The gradient $|\delta\Psi'| \sim 10^{-8} \text{ kpc}^{-1}$ is set by the weak $\Lambda_0 R_{\text{gal}}$ channel, which is 10^4 – $10^6 \times$ below the Vainshtein threshold for any physically reasonable coupling.

5.4 Energy Density Deficit

From equation (15) at the locked-action gradient:

$$\begin{aligned}\rho_{\Psi}(10 \text{ kpc}) &= 1.6 \times 10^{-3} M_{\odot} \text{ kpc}^{-3} \\ \rho_{\text{iso}}(10 \text{ kpc}) &= 4163 M_{\odot} \text{ kpc}^{-3} \\ \frac{\rho_{\text{iso}}}{\rho_{\Psi}} &= 2.7 \times 10^6.\end{aligned}\tag{17}$$

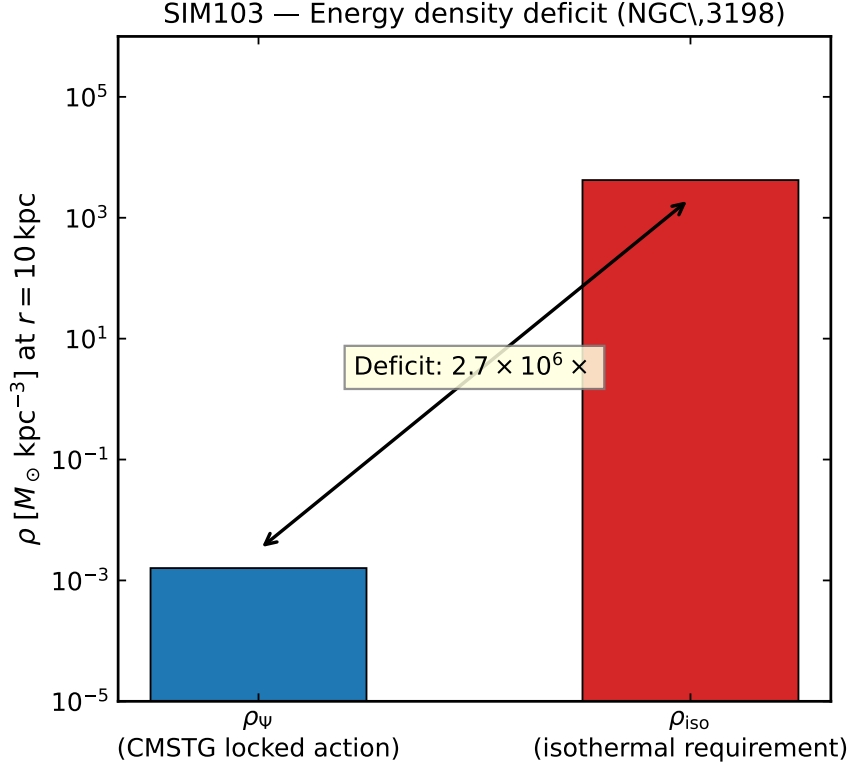


Figure 5: Energy density deficit at galactic scale (SIM103). The locked-action CMSTG scalar density ρ_{Ψ} (blue bar) vs the isothermal halo requirement ρ_{iso} (red bar) at $r = 10 \text{ kpc}$ for NGC 3198 ($v_{\text{flat}} = 157 \text{ km s}^{-1}$). The deficit is $2.7 \times 10^6 \times$. A direct coupling $\beta \approx 4.9 \approx 1,600 \times \Lambda_0$ would be required to supply the missing density, far exceeding the cosmological bound $\Lambda_0 \lesssim 0.095$.

5.5 Direct Coupling Constraint

To close the density gap with a direct matter coupling β , we need $\Lambda_0^{\text{eff}} = \Lambda_0 + \beta \approx 4.9$, requiring $\beta \approx 4.9 \approx 1,600 \times \Lambda_0$. This coupling is not contained in the locked action and would violate cosmological homogeneity constraints: a coupling of order unity would modify G_{eff} by $O(1)$ on cosmological scales, in direct contradiction to the SIM88/SIM90 cosmological consistency tests (Paper II of this series [8]).

6 Combined Constraint

The root cause of all four failures is the single ratio $2\Lambda_0|R_{\text{gal}}|/m_0^2 \approx 5 \times 10^{-7}$: the curvature coupling is $\sim 2 \times 10^6$ times weaker than the field mass on galactic scales (simulation

Table 2: Systematic exhaustion of galactic dark matter mechanisms in CMSTG.

SIM	Mechanism	Deficit	Verdict
SIM99	Static Yukawa $T^{(\Psi)}$	Exponential profile vs $1/r^2$	FAIL
SIM100	Time-domain condensation	$2\Lambda_0 R_{\text{gal}} /m_0^2 \approx 5 \times 10^{-7}$	FAIL
SIM103	Vainshtein kinetic screening	$\mathcal{V} \leq 3.7 \times 10^{-6}$	FAIL
SIM103	Direct matter coupling β	$\beta \sim 1,600\Lambda_0$ required	FAIL
Combined: all mechanisms exhausted at $\Lambda_0 = 0.003$			
Energy density deficit at $r = 10 \text{ kpc}$: $2.7 \times 10^6 \times$			

test mass $m_0 = 0.1 \text{ kpc}^{-1}$; the locked cosmological m_0 gives an even smaller ratio, $\approx 3.6 \times 10^{-8}$). This makes the scalar field cosmological (Compton wavelength $\gg$ galactic size) and prevents any galactic-scale field gradient from building up to isothermal amplitudes.

7 Minimum New Physics

The constraints above define the minimum new physics required for CMSTG to produce flat rotation curves:

Option A: Modified scalar mass. A position-dependent mass $m_0^2 \rightarrow m_0^2(\rho)$ that drops to $m_0^2 \lesssim 2\Lambda_0|R_{\text{gal}}|$ in galactic environments would allow curvature-driven condensation. This is a chameleon-type mechanism [4] with galactic screening. However, any reduction in m_0 inside galaxies must restore $m_0 \sim 10^{-3}H_0$ on cosmological scales to preserve the SIM90/SIM98 consistency (Paper II [8]).

Option B: Direct Ψ -baryon coupling. A coupling $\mathcal{L} \supset \beta\Psi\rho_b$ with $\beta \sim O(1)$ would supply the missing density. This is not derivable from the current locked action and introduces a new free parameter at gravitational strength. It must simultaneously satisfy:

- $|\beta\Psi_{\text{cosmo}}| \ll \Lambda_0$ on cosmological scales (else violates SIM88);
- $\beta + \Lambda_0 \approx 4.9$ on galactic scales.

This requires β to be screened cosmologically and activated galactically—a coupled chameleon architecture.

Option C: New condensation channel. A stable sextic potential $V_6 = +\mu_6\Psi^6$ with $\mu_6 > 0$ would prevent the blow-up of the attractive quartic below λ_{low} , opening a window where the field can form a stable, self-bound condensate. The viability of this requires dedicated simulation (Phase 5 direction).

All three options require new physics not contained in the Phase 1 locked action. The constraint established here is not a quantitative shortfall to be tuned away; it is a structural exclusion of all mechanisms within the current action class.

8 Conclusions

We have established that the Phase 1 canonical locked action of CMSTG cannot replace cold dark matter at galactic scales through any of the mechanisms accessible within the action class. The three independent simulation programmes establish:

1. **Static profile (SIM99):** The Yukawa field profile is exponentially steeper than the isothermal halo requirement. The quartic potential modifies the central amplitude but not the exponential radial dependence.
2. **Time-domain dynamics (SIM100):** The curvature coupling $2\Lambda_0|R_{\text{gal}}|$ is $\sim 2 \times 10^6$ times weaker than m_0^2 at all galactic densities (simulation test mass; locked m_0 is weaker still). Condensation requires neutron-star densities. No quartic self-coupling produces a flat-curve condensate: the sub-threshold window gives Yukawa profiles; the super-threshold window gives runaway blow-up.
3. **Vainshtein screening (SIM103):** The Vainshtein ratio $\mathcal{V} \leq 3.7 \times 10^{-6}$ across the full (α_V, M_V) parameter space. The field gradient is 10^4 – $10^6 \times$ below the nonlinear threshold at all galactic radii.

The energy density deficit at $r = 10$ kpc is $2.7 \times 10^6 \times$ under all three mechanisms. A direct matter coupling $\beta \sim 1,600 \Lambda_0$ would supply the missing density but is excluded by cosmological constraints (Paper II [8]).

This is a clean negative result: every mechanism within the CMSTG action class has been tested, every mechanism fails, and the reason is a single structural ratio ($2\Lambda_0|R_{\text{gal}}|/m_0^2 \approx 5 \times 10^{-7}$ at the simulation test mass, even smaller at the locked cosmological m_0) rather than a parameter choice. A first-principles coupling between Ψ and baryonic matter, with appropriate cosmological screening, is the minimum new physics required for galactic-scale predictions.

Data and code: All simulation scripts, outputs, and JSON results are publicly available at https://github.com/cisomorph/Curvature-Memory_Scalar-Tensor_Gravity (simulations SIM99, SIM100, SIM103).

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